

CSE 4502: Operating System Lab

Lab 1: The Command Line Essentials

Jibon Naher, Lecturer, CSE

February 26, 2026

Introduction

This assignment is designed to provide essential hands-on experience with **20 fundamental commands** used for creating, viewing, manipulating, and managing files and directories in a Ubuntu (Linux) environment. All commands must be executed within your system's terminal.

1 Part 1: Preparation & Setup (4 Commands)

Objective

Create a dedicated workspace and practice navigating the file system.

Tasks

- Create a primary directory named `linux_assignment`.
- Navigate into this new directory.
- Create a sub-directory named `Documents`.

Table 1: Commands for Setup

No.	Command	Description
1	<code>pwd</code>	Print Working Directory (Verify your current location).
2	<code>mkdir</code>	Make Directory (Create <code>linux_assignment</code> and <code>Documents</code>).
3	<code>cd</code>	Change Directory (Move into <code>linux_assignment</code> and navigate between folders).
4	<code>ls</code>	List directory contents (Use <code>ls -l</code> to see file permissions and details).

2 Part 2: File Creation and Viewing (5 Commands)

Objective

Create several files and practice viewing their contents and metadata efficiently.

Tasks

- Create two empty files: `report.txt` and `notes.log`.
- Add a line of text (e.g., "Experiment data collected on 2024-01-01") to a third file, `data.csv`, using a single command line redirection.
- Create a large, empty file (e.g., 10MB) to practice paging.

Table 2: Commands for Creation and Viewing

No.	Command	Description
5	<code>touch</code>	Create a new, empty file or update the timestamp of an existing one.
6	<code>echo</code>	Output text (Used with redirection <code>></code> to populate <code>data.csv</code>).
7	<code>cat</code>	Concatenate and display the entire file content (View <code>data.csv</code>).
8	<code>file</code>	Determine the file type (Check <code>report.txt</code>).
9	<code>less</code>	Pager utility to view files one page at a time (Use for the large file).

3 Part 3: Copying, Moving, and Renaming (4 Commands)

Objective

Practice moving and copying files/directories and renaming objects.

Tasks

- Copy `report.txt` to the `Documents` sub-directory.
- Rename `notes.log` to `server_logs.txt` in the current directory.
- Move `data.csv` into the `Documents` sub-directory.
- Create a temporary empty directory called `temp` and immediately remove it.

Table 3: Commands for Manipulation

No.	Command	Description
10	<code>cp</code>	Copy files.
11	<code>mv</code>	Move or rename files and directories.
12	<code>cp -r</code>	Copy directories recursively (Create a new directory <code>backup</code> and copy <code>Documents</code> into it).
13	<code>rmdir</code>	Remove an empty directory.

4 Part 4: Searching and Permissions (5 Commands)

Objective

Locate files, search for content, manage file sizes, and handle access permissions.

Tasks

- Search for the location of `server_logs.txt` starting from your home directory.
- Search for the text "Experiment" *within* `Documents/data.csv`.
- Check the total disk space used by the `linux_assignment` directory in human-readable format.
- Change the permissions of `server_logs.txt` using the numeric mode 600 (read/write only for the user).

Table 4: Commands for Searching, Size, and Permissions

No.	Command	Description
14	<code>find</code>	Search for files in a directory hierarchy based on criteria.
15	<code>grep</code>	Search for a specific text pattern within file content.
16	<code>chmod</code>	Change file access permissions.
17	<code>chown</code>	Change file owner and/or group.
18	<code>du</code>	Disk Usage (Check the size of a directory in human-readable format, <code>du -h</code>).

5 Part 5: Advanced Viewing and Cleanup (2 Commands)

Objective

View partial file content and remove the entire assignment structure.

Tasks

- View only the first 5 lines of `server_logs.txt`.
- Remove the entire `linux_assignment` directory, including all sub-directories (`Documents`, `backup`) and files.

Table 5: Commands for Final Steps

No.	Command	Description
19	<code>head</code>	Output the beginning (first part) of files.
20	<code>rm</code>	Remove files or directories (Use the recursive option, <code>rm -r</code> , for directory removal).

Submission Requirements

1. **List of Commands:** A numbered list of all 20 commands used (matching the numbers in the tables).

2. **Syntax:** The exact command line syntax you used for each command (e.g., `touch report.txt`).
3. **Execution Proof:** A single, clear, scrolling screenshot (or multiple combined screenshots) showing the execution and output of **all 20 commands** performed sequentially in your terminal. This demonstrates successful completion of the entire assignment flow.